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(1) Given

Hexagonal cellular network, $A = 1,200\ km^2$

Cell radius, $R = 1.8\ km$

available spectrum, $S = 48\ MHz$

Channel $B = 200\ kHz$

Cluster size $N_1 = 7$

$N_2 = 4$

Reduction = 15%.

i) Area of Hexagonal cellular network

$$A = \frac{3\sqrt{3}}{2}(R^2)$$

$$= \frac{3\sqrt{3}}{2}(1.8)^2$$

$$= \frac{3\sqrt{3}}{2}$$

$$= 8.4\ km^2$$

Total cells

$$\frac{1200}{8.4} = 143 \quad = 142.85 \text{ (approx)}$$

Total available channels

240
48000
200
1

= 240

Cluster size $N_1 = 7$ $N_2 = 4$

240 240

$N_1 = \frac{240}{7}$ $= \frac{240}{4}$ 60

= 34.2 = 60

Formula :

$$D = R\sqrt{3}N$$

$= 1.8(\sqrt{3})(34.2)$

$= 106.62$

$$Q = \sqrt{3}N$$

$= \sqrt{3}(3.6)$

$= 6.2354.$

∴ Total cells = 142.85

Available channels = 240

2) PLMN time = 40 mints

Cell boundaries - 15

Speed = 90 km/h

PLMN handoff = 120 ms

Nss path = 80 ms

Authentication = 350 ms

roaming = 250 ms

transitions = 2

fails = 3

Normal Handoff

15 (120+80)

= 3000 ms

= 3S

Recovery

3 x 1000 = 1.2 $

Roaming =

2(350 + 250)

= 2(600)

$= 1.2\,S$

Total interption $= 3 + 1.2 + 1.2$

$= 5.4$

Data delay

512×5.4

$= 2764.8$

$$\frac{2764.8}{8}$$

$= 345.6$

Data Delay = 345.6 mb.

Total interption = 5.4

Normal handoff = 3S

Recovery = 1.2 S

Roaming = 1.2 S

(3) i) B Given

BTS = 12

Sectors = 3

Transceivers per second = 4

TDMA Timeslots = 8

Time per sector = 2

Subscribers = 1,200

Calls per hour = 1.2

Time = 90

Channels = 16 kbps

i) Calculate traffic channal per sector

$$= \frac{12 \times 100}{3}$$

for BTS = 400

for BSC = $\frac{4 \times 12}{3}$

$$= 16$$

ii) Time per sector

$$= 8 \times 200$$

$$= 1600 \text{ s}$$

minimum no. of TRXs

$$= \frac{12000 \times 1.2}{9\phi}$$

$$= 160$$

(3ii) Given, FDMA

$f = 30\text{kHz}$

$N = 7$

TDMA $f = 200\text{kHz}$

users = 8

$N = 4$

CDMA

Uses = 1.25MHz

Chip Y = 1.2288 Mcps

Voice data rate = 9.6 kbps

$$\text{FDMA} = \frac{30000}{7}$$

$$= 1.53$$

$$\text{TDMA} = \frac{2000 \times 100}{4 \times 1}$$

$$= 500$$

(4) Given, No. of files = 240

Size = 18 MB

Timeslots = 4

Rate = 20 kbps

Capacity = 18 %.

Throughput = 85 %.

Packet transmission rate = 12 %.

mobility interruptions = 8

time = 2.5 seconds

i) Calculate aggregate data rate

$$= \frac{20 \times 240}{18}$$

$$= 14.8148$$

ii) GPRS throughput = $\frac{85 \times 4}{100}$

$$= 0.03$$

iii) Total data volume

$$= \frac{18 \times 18}{4}$$

$$= 81$$

Calculate the time

$= 240 \times 4$

$= 960$ sec

GPRS. = 16 hours.

UMTS transfer time

$= 16 \times 384$

$= 6144$ S

By comparing the both GPRS and UMTS Timeslots are 960 S and 6144 S.

(5) Given

Software update = 120GB

No. devices = 2000

GSM provides = 9.6 kbps

GPRS provides = 64kbps

retransmission = 10%.

5G provides = 25 Mbps

P. = 4%.

Devices = 40.

i) Calculate data volume per devices

$$= \frac{\overset{60}{\cancel{120}} \times 1000}{\cancel{2000}}$$

$$= 60 \text{ MB/per}$$

Convert it into bits

$$= 60 \times 8$$

$$= 480 \text{ Mb/pd.}$$

ii) Individual transfer times

$$= \frac{2000}{480}$$

$$= 4.166 \text{ \$}$$

Convert into hours

$$= \frac{4166}{3600}$$

$$= 13.87 \text{ hours.}$$

iii) In GSM

64kbps

$$= 120 \times 1.10$$

$$= 132$$

$$GSM = \frac{132}{64}$$

$$= 2.062$$

In 5G

25Mbps

$$= \frac{250}{100}$$

$$= 2.5 = 60$$

In GPRS

$$= \frac{64}{40} = 1.60$$

iv) Calculating the percentage of redution

$$\frac{T_{GPRS} - T_{5G}}{T_{GPRS}}$$

$$= \frac{1600000 - 60}{1600000}$$

$$= 99.999\%$$

percentage reduction in completion time achived by 5G compared with GPRS is 9.999%.